

Capstone Impact Report

1. Purpose

This report measures the impact of right-sizing the agentic engineering pipeline and records the evidence available before the final production-like capstone run.

The report separates measured results from metrics that still require final runtime execution. No missing metric is estimated or invented.

2. Measured Deterministic Conversion

One stable workflow step, handoff validation, was converted from agentic execution to deterministic code.

The measurements recorded in docs/calibration-log.md are:

Metric	Before: Agentic Validation	After: Deterministic Validation	Result
Average cycle time	45 seconds	0.2 seconds	Improved
Token cost per run	\$0.003	\$0	Improved
Review latency	~30 seconds	~5 seconds	Improved
Deterministic harness checks	7/7 passing	7/7 passing	Quality preserved
Output variance	Not recorded	Zero across 3 runs	Repeatable

2A. Required Five-Metric Before/After Summary

The following table consolidates the five impact metrics required for the capstone. Values are reported only where supporting evidence exists; unsupported values are explicitly marked rather than estimated.

Required Metric	Before	After	Evidence / Interpretation
Quality	7/7 deterministic checks passing	7/7 deterministic checks passing; later integrated regression suite 24/24 passing	Measured quality was preserved while the validation step was simplified.
Review latency	~30 seconds	~5 seconds	Measured for the handoff-validation path.
Defect rate	No reliable baseline defect-rate percentage recorded	No general defect-rate percentage claimed; final successful run recorded 0 rejected evaluation items across 8 recorded review items	Evidence shows a successful evaluated run, but is insufficient to claim a general 0% defect rate.

Required Metric	Before	After	Evidence / Interpretation
Cycle time	45 seconds	0.2 seconds	Measured for handoff validation after conversion from agentic to deterministic execution.
Cost	~\$0.003 model-token cost per validation run	\$0 model-token cost for the deterministic validation step	This comparison applies only to the converted validation step, not the complete multi-agent workflow.

This comparison shows that the stable validation step became faster and removed its model-token cost without reducing the recorded deterministic quality result. Where a trustworthy baseline was unavailable, the report intentionally avoids inventing a value.

4. Cost Impact

The agentic handoff-validation step recorded a token cost of approximately:

\$0.003 per run

The deterministic implementation records:

\$0 per run

for model-token cost.

The deterministic implementation therefore removes model-token cost from this specific validation step.

This does not mean the complete multi-agent workflow has zero cost. Other agentic steps may still require model execution.

5. Review-Latency Impact

Recorded review latency decreased from approximately:

30 seconds

to:

5 seconds

for the converted validation path.

This is a measured improvement for the converted step, not yet a claim about total end-to-end capstone review latency.

6. Reliability and Repeatability

Before conversion, the recorded deterministic harness result was:

7/7 passing

After conversion, the result remained:

7/7 passing

The deterministic implementation also produced zero output variance across three repeated runs, with no differences reported.

This provides evidence that the conversion improved speed and cost without reducing the measured structural quality of the step.

7. Integrated Regression Evidence

After the deterministic validator was integrated into the larger workflow, the calibration log recorded:

24/24 harness checks passing

The policy suite also passed.

This indicates that the conversion did not introduce a detected regression in the recorded integration check.

8. Governance Impact

Calibration identified several governance near-misses:

- overly broad Implementer delete access;
- Reviewer access that could trigger an inappropriate testing capability;
- retrieval above the Implementer's classification ceiling.

The final governance design responds to these risks through:

- role-specific MCP allow-lists;
- data-classification ceilings;
- read-only role boundaries;
- policy tests;
- governed-file tests;
- red-team testing;
- human checkpoints.

These controls reduce the blast radius of an incorrect or overreaching agent action.

9. Final Production-Like Run Evidence

The final production-like capstone run was completed using the course-approved Claude Haiku 4.5 model.

Metric	Result
Model	anthropic/claude-haiku-4.5
Agent path	planner â†’ implementer â†’ reviewer â†’ tester
Full-pipeline runtime	59.6 seconds
Audit entries	15
Transcript	logs/capstone-final-run-002.json
Audit log	logs/capstone-final-run-002.log

The run successfully exercised retrieval, storage reads, and an implementer storage write. Reviewer and tester steps completed with recorded validation results.

The final run establishes production-like execution evidence. Full-pipeline quality score, defect-rate measurement, and reliable cost-per-run measurement are still pending because the current transcript does not provide a trustworthy final

cost value.

10. Baseline Limitation

The repository contains historical Module 1 PRD, rubric, and iteration evidence, but the currently available Module 1 artifacts do not provide a complete single baseline containing every capstone impact metric.

Historical measurements must therefore be labeled according to their actual source rather than reconstructed or estimated.

The final capstone comparison will distinguish:

1. historical Module 1 evidence;
2. deterministic-conversion measurements;
3. final production-like capstone measurements.

11. Tool-Evolution Drill

A controlled permission-change drill was performed on the storage MCP allow-list.

The documentation-writer role temporarily lost its granted `read_entry` permission.

Results:

Stage	Policy Test Result
Baseline	6 passed
Permission revoked	1 failed, 5 passed
Permission restored	6 passed

The failing test was:

`test_documentation_writer_policy_matches_enforcement`

This demonstrated that the evaluation and governance system can detect permission drift between the documented policy and executable MCP enforcement.

The permission was restored and the same policy suite returned to a fully passing state.

Detailed evidence is available in:

`docs/capstone-tool-evolution-drill.md`

12. Final Conclusion

The available evidence supports measurable improvements from right-sizing and governing the capstone workflow.

For the handoff-validation step:

- average cycle time decreased from 45 seconds to 0.2 seconds;
- model-token cost decreased from \$0.003 per run to \$0 for that deterministic step;
- review latency decreased from approximately 30 seconds to approximately 5 seconds;
- deterministic harness quality remained 7/7 passing before and after conversion;
- repeated deterministic output showed zero variance across three runs;
- the integrated regression suite remained 24/24 passing.

The final production-like workflow also completed the expected planner → implementer → reviewer → tester path in 59.6 seconds. It exercised retrieval, persistent-storage reads, an Implementer storage write, downstream Reviewer verification, and Tester validation.

In the successful final run, all recorded evaluation items were approved: 3 Planner review items, 4 Reviewer review items, and 1 Tester review item. No recorded review item was rejected in that run. This is evidence of a successful evaluated run, but it is not sufficient to claim a general 0% defect rate across all workloads.

Governance enforcement was also demonstrated separately. A documentation-writer attempt to call the unauthorized `write_entry` operation was denied by the storage MCP allow-list and recorded as `authorization_denied` in the storage audit log, while the role's permitted `read_entry` operation succeeded.

The tool-evolution drill further demonstrated that policy drift is detectable: the policy suite changed from 6 passing tests to 1 failing and 5 passing when a permission was intentionally revoked, then returned to 6 passing after the permission was restored.

Two limitations remain explicit. The final transcript records a zero token-cost field, but the available evidence does not establish a trustworthy full-pipeline model cost, so no unsupported end-to-end cost claim is made. Human-checkpoint conditions are defined in governance policy, but the successful final production-like run did not require or trigger human escalation.

Overall, the evidence shows that deterministic conversion reduced latency and model-token cost for a stable step without detected quality regression, while scoped MCP permissions, evaluation gates, audit logging, and CI policy checks improved the workflow's operational control and traceability.